

Internship Progress Report

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Internship Topic: Design and Development of Just-in-Time Immersive Video Coaching to Support CPAP Adherence

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1. Project Context

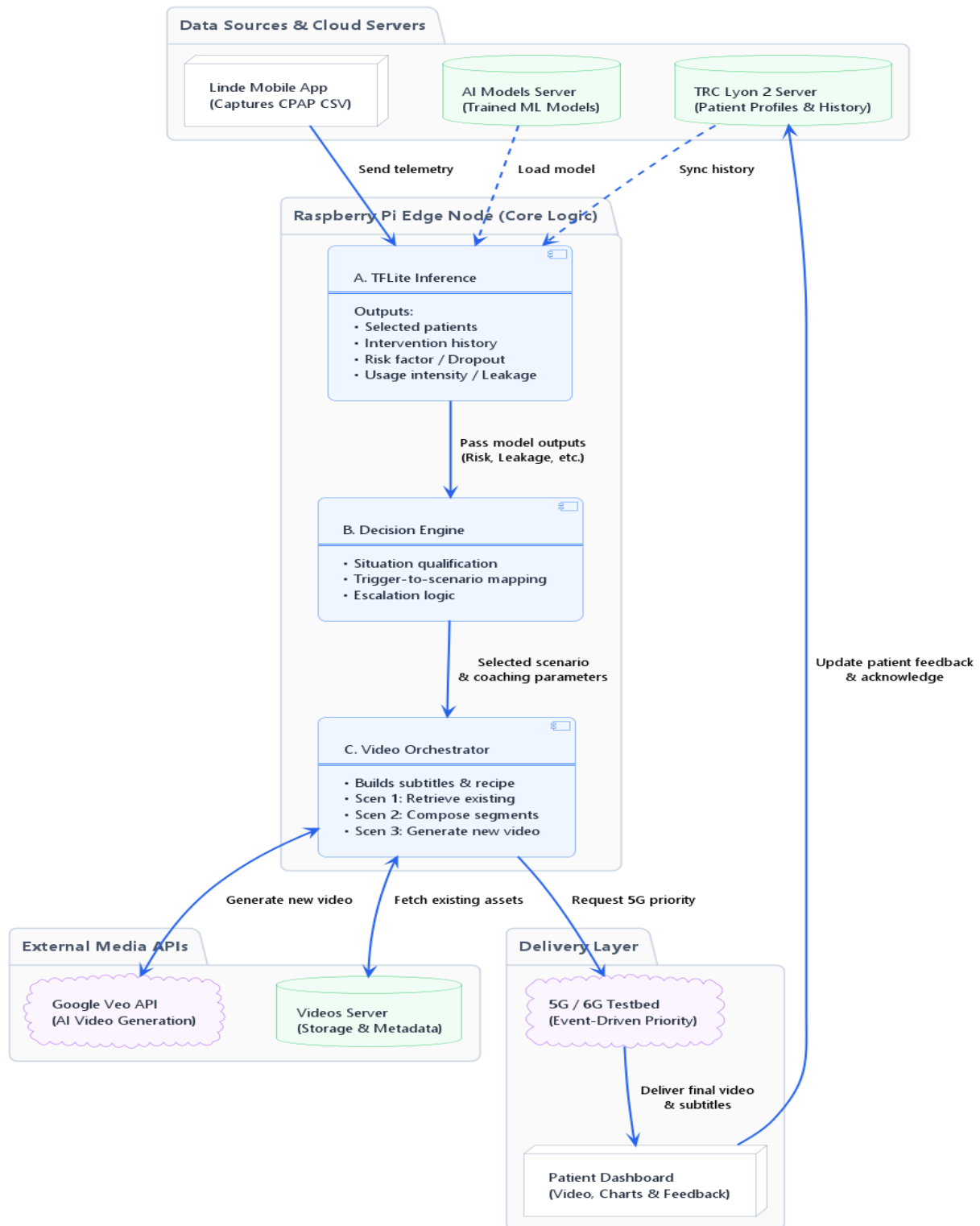
Design and implementation of a 5G Quality on Demand (QoD) enabled dynamic video generation pipeline. The project utilizes Edge Computing (MEC) via Raspberry Pi to detect real-time CPAP telemetry anomalies. Upon detection, it automatically orchestrates the generation and delivery of personalized, medically accurate coaching videos to patients under strict 5G/6G network delivery constraints (<60 seconds).

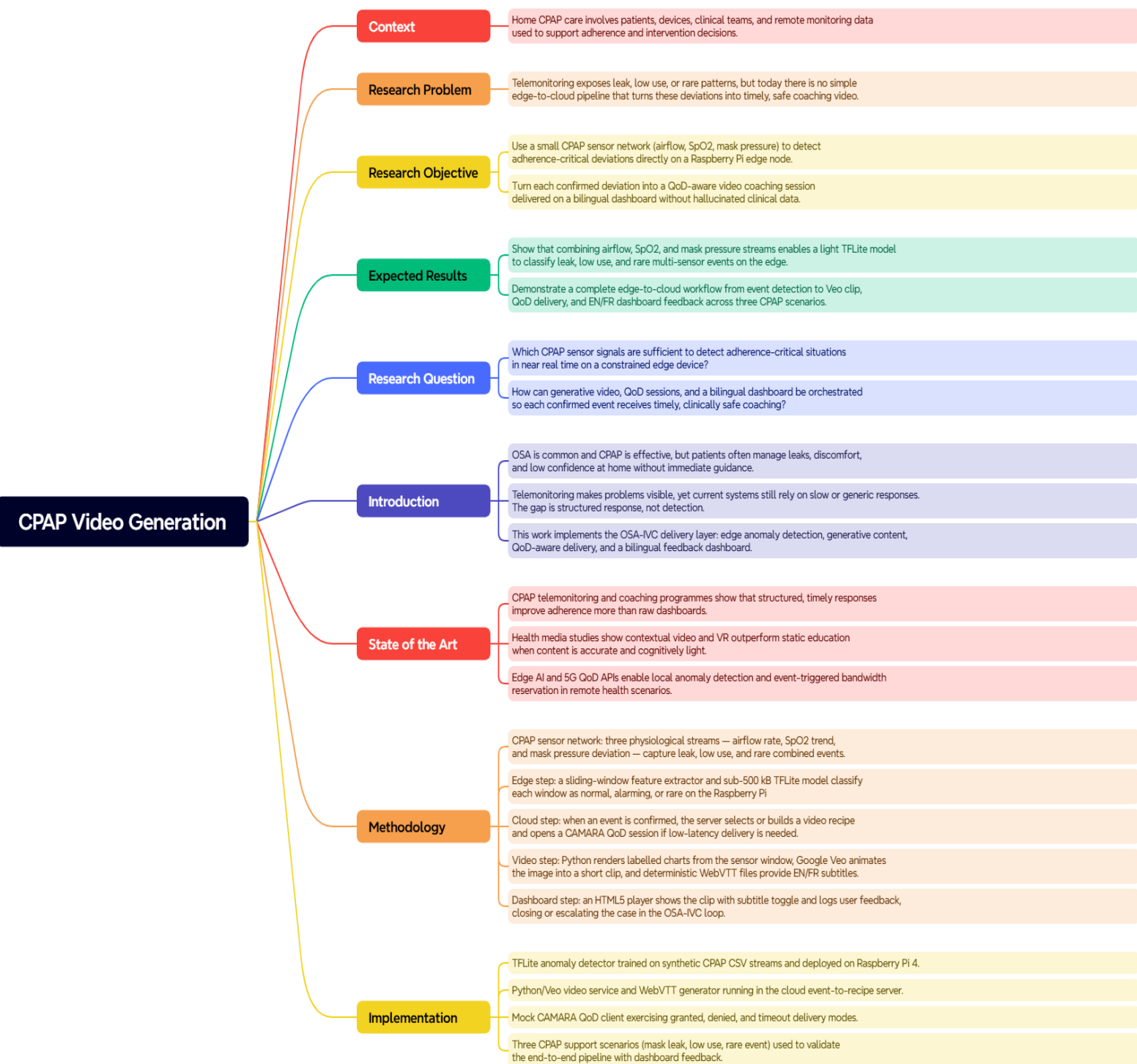
2. Summary of Achievements to Date

During this second month, I focused on formalizing the orchestration logic, building out the offline video generation scripts, and writing the academic conference paper while awaiting the final machine learning model and cloud API credentials.

- **Video Coaching Scenarios & Orchestration Logic (Week 6 & 7):**
 - Finalized the orchestration logic mapping specific telemetry events to video delivery paths, in preparation for the incoming TFLite model.
 - Defined the three primary coaching delivery scenarios: *Retrieve existing assets*, *Adapt existing assets* (segment composition), and *Generate new video* (for complex multi-sensor or biomarker anomalies).
 - Developed the offline "Generative Fallback" pipeline: The Python backend now successfully generates deterministic, medically accurate static charts to prevent AI hallucination, paired with dynamically generated French and English HTML5 WebVTT (.vtt) subtitles.
- **Academic Conference Paper (Week 8):**
 - Created a Mindmap also authored and iteratively refined a 4-page A4 double-column conference paper for the IEEE EMBS BSN (Body Sensor Networks) 2026 conference.
 - The paper, titled "Edge-to-Cloud Video Coaching for CPAP Home Support," details the data-to-impact pipeline, the edge classifier logic, and the CAMARA QoD network bandwidth reservation strategy.

- Incorporated supervisor feedback to explicitly connect the three video generation paths and emphasize the necessity of the extended biomarker data layer.
- Finalized the LaTeX formatting, optimizing the text to strictly fit within the 4-page limit while retaining comprehensive technical detail.





Edge-to-Cloud Video Coaching for CPAP Home Support: TFLite Anomaly Detection, Generative Content, and QoD-Aware Delivery

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Abstract—Continuous positive airway pressure (CPAP) telemonitoring reveals issues such as mask leaks, low nightly usage, and residual events, yet non-adherence and therapy dropout remain persistent challenges. This paper introduces a data-to-impact pipeline that goes beyond CPAP telemetry by incorporating digital biomarkers and self-reported symptoms, targeting a reduction in dropout events by 20% and a 30 minute per night increase in nightly use for at-risk OSA patients. It connects this information to personalized video coaching through two sequential models. A multimodal deviation detection model, running as a quantized TFLite classifier on a Raspberry Pi, identifies abnormal or risky patient situations. A deviation to video model converts the detected issue, severity, and patient context into coaching content. Clips are created via a structured workflow where Python charts capture clinical data and Google Veeo provides animations. Delivery occurs during a CAMARA Quality-on-Demand (QoD) session targeting 95% activation success and end-to-end latency under 60 seconds for cached assets. A bilingual web dashboard presents the video with English and French subtitles. Tests across three representative support cases confirm end-to-end pipeline completion and lay the groundwork for prospective medical validation.

Index Terms—CPAP adherence, OSA, edge AI, TensorFlow Lite, Quality-on-Demand, CAMARA API, generative video, VTT, telehealth, digital biomarkers, patient empowerment.

I. INTRODUCTION

Continuous positive airway pressure is the standard treatment for moderate-to-severe obstructive sleep apnea (OSA), but sticking to the treatment at home is challenging [1], [2]. Modern telemonitoring platforms allow clinical teams to access data on usage patterns, leaks, and residual events [2], [5]. Yet, adherence only improves when monitoring is part of a structured response system. Passive data channels typically

offer generic or delayed feedback, which seldom leads to lasting behavior changes [3], [4]. A key drawback of current CPAP telemonitoring is that device data only capture the treatment aspect. They can indicate a leaking mask or low usage but do not clarify why a patient is having difficulty, how they feel, or what kind of support would help the most. Our prior work established the case for linking CPAP telemetry to structured coaching responses, but two limitations remained unresolved. First, the intervention models operated on device metrics alone and could not generate differentiated actions for patients sharing the same deviation class but differing in symptom history, fatigue levels, or prior coaching outcomes. Second, there was no mechanism to identify why a situation failed to improve after an earlier intervention, limiting the system's ability to escalate or adapt support for specific at-risk subgroups. To empower patients to manage their therapy, the system needs a broader understanding of their home situation, combining device telemetry with digital biomarkers and self-reported symptoms. In practice, three scenarios lead to the greatest challenges: ongoing mask leaks, low nightly usage, and combined changes across multiple sensor data streams. In these instances, understanding the root cause is as critical as the deviation itself.

This paper proposes a data-to-impact pipeline linking various patient-side sensing to personalized video coaching. The pipeline consists of two sequential models. A multimodal deviation detection model identifies risky or abnormal patient situations from the combined data layer. A Deviation to Video model then converts the identified deviation, its severity, and patient context into a coaching action. The four contributions

are: (i) a patient-side data layer that enhances CPAP telemetry with digital biomarkers, self-reported symptoms, and feedback after previous coaching, (ii) a multimodal deviation detection model (TFLite, Raspberry Pi) that classifies diverse sensor data into actionable deviation categories, (iii) a Deviation to Video model with a structured workflow where Python charts encode all clinical values and Google Veo animates the output, and (iv) a QoD-aware delivery system and bilingual web dashboard for patient empowerment.

II. STATE OF THE ART

Four research areas drive the design: CPAP telemonitoring and adherence, health video and patient engagement, remote digital health and multimodal sensing, and edge AI with QoD networking.

A. CPAP Adherence and Telemonitoring

Telemonitoring gives medical teams data on usage, leaks, and residual events between visits [2], [5]. Programs that integrate telemonitoring within a defined response pathway show better adherence than passive monitoring alone [4], [6]. However, pathways focusing solely on CPAP have limited effectiveness because device data do not capture the patient's subjective experiences, symptoms, or larger health context. Expanding the data layer with patient-reported outcomes and digital biomarkers is essential for moving from mere monitoring to real patient empowerment [7].

B. Health Video and Patient Engagement

Research on 360° video and virtual reality in healthcare shows that how useful something seems relies on its relevance, immersion, and guidance [9]. In patient education, videos that visualize the patient's own data and offer clear, step-by-step explanations are more likely to aid self-management than generic clips [9], [10]. These principles inform the design of the programmatically structured, chart-based videos.

C. Remote Digital Health, Multimodal Sensing, and Edge AI

Large-scale remote monitoring deployments show that sensing is only valuable when connected to structured response workflows [5]. Combining device telemetry with physiological signals and patient-reported data improves our understanding of the patient's situation [7]. TFLite quantized models can achieve full-precision accuracy, making Raspberry Pi a suitable platform for local analysis of physiological time series [12]. The CAMARA QoD API allows for on-demand bandwidth reservation for each event instead of continuous reservation [13], [14].

III. SYSTEM ARCHITECTURE

The system is set up as a data-to-impact pipeline that changes various patient-side evidence into personalized video coaching. Fig. 1 displays four components: (A) a Raspberry Pi edge node that collects CPAP sensor data, digital biomarkers, and patient feedback while running the TFLite deviation-detection model, (B) a cloud orchestration layer that connects confirmed events to coaching paths, manages the video

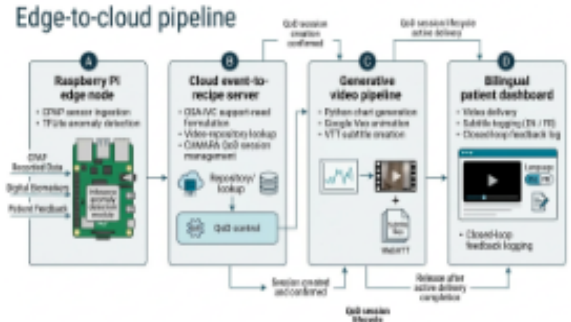


Fig. 1. Data-to-impact pipeline for CPAP video coaching.

database, and controls the CAMARA QoD session, (C) a generative video pipeline that produces a Python chart, a Google Veo animation, and bilingual VTT subtitle files, and (D) a bilingual patient dashboard that presents the video and captures feedback logs. Each component has a clear interface, allowing models, media services, or network features to be updated separately.

A. Patient Side Data Layer

A key contribution is expanding the sensing layer beyond just CPAP device telemetry. Data from CPAP alone, like usage duration and mask leak, can identify basic treatment issues but do not capture the patient's complete situation at home. A patient might have normal device readings but still feel discomfort, fatigue, or declining confidence, which could later hurt adherence, and high leak levels alone do not differentiate between mask-fit issues or problems from body position. The data layer, therefore, combines four complementary sources. Digital biomarkers like SpO2 and heart rate variability complement usage data and can signal early signs of worsening therapy control. Self-reported symptoms gathered through the dashboard, such as fatigue, headaches, throat dryness, and sleep quality, provide insights that device data alone cannot offer. Feedback from patients about previous coaching helps assess whether a particular action addressed the problem. Together, these sources enable the system to distinguish, for instance, whether a patient has leaks and reports discomfort, or leaks but feels the device is working well, and to understand why a previous video did not resolve the issue.

B. Edge Node - Multimodal Deviation Detection

The edge node is a Raspberry Pi 4 that runs the first model in the pipeline, a multimodal deviation detection model that classifies the current patient situation based on the combined data layer. Raw sensor data stays on the device to reduce bandwidth use and privacy risks, and only aggregated event data is sent to the cloud. Three structured CPAP sensor channels are collected: airflow rate, oxygen saturation trend, and mask pressure variation. Each channel is separated into 20-second windows sampled at 1 Hz. For each window, the feature

TABLE I
TFLITE EDGE MODEL CONFIGURATION

Parameter	Value
Architecture	Multi-layer perceptron (MLP)
Input features	36 (12 per sensor channel)
Window size	20s at 1 Hz
Output classes	3 (Normal / Alarming / Digital biomarker)
Quantisation	INT8 post-training
Model size (TFLite)	<500kB
Training data	CPAP-structured, synthetic anomaly injection
Target hardware	Raspberry Pi 4 (ARM Cortex-A72)

extractor calculates mean, variance, maximum excursion, and zero-crossing rate per channel, creating a 36-dimensional input vector. This vector is fed into a quantized TFLite multi-layer perceptron, summarized in Table I, that categorizes the window as Normal, Alarming, or Digital-biomarker. Sustained mask leaks or significant drops in oxygen saturation trigger an alarm. Linked deviations across multiple channels over a sustained period are identified as a digital-biomarker event. If the classifier's confidence for a non-normal class goes beyond the operating threshold, the edge node prepares a compact event package and sends it to the cloud using a secured REST connection.

C. Cloud Orchestration - Deviation to Video Model

The cloud orchestration layer implements the second model, a Deviation to Video model that converts confirmed deviations, their severity, and patient context into a video recipe. This model functions as a rule-and-retrieval system applying clinical logic to choose the coaching action. It recognizes three paths for video creation. In the first, a validated repository asset matching the event type and patient context is retrieved directly. In the second, an existing asset is modified by updating its chart data, threshold notes, and subtitle content to reflect the patient's current state. In the third, if there is no suitable asset, a full programmatic generation workflow produces a new clip that is saved for future use. The layer carries out three tasks. First, it connects the event to an intervention path using the deviation class, severity, and available context. A mask-leak event leads to a mask-fit coaching path, a low-use event goes to a motivation-focused route enhanced with usage trends and recent feedback, and a digital-biomarker event routes to an escalation path for technician review. Second, it searches the video repository, if a suitable clip exists, it is retrieved or modified. Third, if no asset is found, the layer triggers the programmatic video generation workflow and opens a CAMARA QoD session.

D. Programmatically Anchored Video Pipeline

The video pipeline serves as the implementation layer of the Deviation to Video model. It separates factual clinical content from visual animation to prevent the generative model from introducing medical errors [11]. A Python script gathers the sensor window data and clinician-defined thresholds for

the confirmed event and creates a full HD 16:9 dark-mode Matplotlib chart showing sensor traces, threshold lines, and marked anomalous segments. All clinical values, colours, and labels are set through deterministic code. This static chart is uploaded to the Google Veo Vertex API as an image-to-video prompt that only specifies camera movement, pacing, and style. Veo produces an animated clip lasting 15 to 30 seconds. A separate script generates VTT subtitle files in English and French using structured narration synced with video timestamps, and both the clip and subtitles are indexed in the content repository for reuse.

E. Bilingual Patient Dashboard

Patients access coaching through a web application. When a coaching video is ready, the dashboard displays it with both English and French subtitles and allows users to switch languages and replay the video without stopping playback, keeping on-screen controls simple to lessen cognitive load. After watching the video, a follow-up prompt captures whether the patient thinks the problem is resolved, determining the case outcome: resolved, repeat, or escalate to technician review. These feedback signals are saved for future retraining of both models.

IV. IMPLEMENTATION AND EVALUATION

A. Model Training and Data

The deviation-detection model is trained on CPAP-styled data streams that fit the statistical characteristics of exported device data, with added anomaly patterns representing three deviation classes [5], [7]. Alarming cases include high leak and temporary oxygen drops, while digital-biomarker cases show multi-channel deviations that might signal worsening conditions. The model is initially trained in full precision and then converted to INT8 TFLite for edge deployment, and the Deviation to Video model is set up as a rule set based on the three coaching paths and clinician-approved threshold configurations [12].

B. End-to-End Scenarios

Three representative support cases test the entire pipeline: a mask-leak case, a low-use case, and a digital-biomarker case. A test harness sends predefined classification sequences to the edge component to trigger each deviation class, and for every case, both QoD grant and fallback are assessed. Table II outlines completion for each stage.

All stages perform as expected across the three cases. In the leak case, a validated repository asset is retrieved directly. In the low-use case, an existing clip is adapted with updated chart data and subtitles. In the digital-biomarker case, a new clip is generated and stored. Language switching works without pausing playback, and the follow-up prompt routes cases to closure or escalation. Visual checks confirm that no clinical values appear in the Veo layer. Table III reports quantitative KPIs measured across all three scenarios.

TABLE II
PIPELINE COMPLETION PER SUPPORT CASE (✓ – PASS, – – NOT APPLICABLE)

Stage	Leak	Low use	Digital biomarker
Edge detection	✓	✓	✓
Event-to-recipe mapping	✓	✓	✓
Repository lookup	✓(retrieve)	✓(adapt)	–
Video generation	–	–	✓
VTT EN + FR	✓	✓	✓
QoD granted	✓	✓	✓
QoD fallback	✓	✓	✓
Web app delivery	✓	✓	✓
Feedback / escalation	✓(closed)	✓(closed)	✓(esc.)

TABLE III
TECHNICAL KPI RESULTS ACROSS SCENARIOS

KPI	Target	Observed
End-to-end delay (cached asset)	≤60 s	<60 s ✓
End-to-end delay (generated clip)	≤15 min	<15 min ✓
QoD activation success rate	≥95%	3/3 sessions ✓
Delivery reliability	≥95%	3/3 cases ✓
Playback stall ratio	≤2%	0% ✓
Edge resource saving (vs cloud)	≥20%	cache path ✓

V. DISCUSSION

The main contribution is the coordinated chain. This chain uses a richer patient-side data layer to improve deviation detection beyond just CPAP metrics. The Deviation to Video model transforms this understanding into targeted coaching. To truly empower patients, it is necessary to extend the data layer with digital biomarkers, self-reported symptoms, and feedback on previous interventions, rather than just generating simple alarms. As device exports and patient responses grow, the dashboard collects feedback that aids in retraining and recalibrating both models based on real-world patterns. This feedback also helps explain why a situation may not have improved after the initial video. The video design, which is programmatically anchored, maintains a clear separation between verified clinical content and generative animation. Selective QoD reservation occurs only on confirmed events, preventing permanent resource allocation while still allowing for timely delivery of crucial coaching when adherence is critical.

VI. CONCLUSION

This paper introduced a data-to-impact pipeline for personalized CPAP video coaching. It addresses three key clinical scenarios: mask leak, low nightly use, and digital-biomarker events. The main contribution is a complete chain that starts with varied patient-side sensing and goes through multimodal deviation detection and Deviation to Video translation, ending with QoD-aware delivery and bilingual dashboard interaction. Relying on CPAP telemetry alone is not enough to

empower patients. By extending the data layer to include digital biomarkers, symptoms, interaction records, and feedback from previous coaching, it becomes possible to gain better awareness of situations and continuously improve models. The pipeline functions correctly across all three support cases, providing a reusable prototype for clinical validation and potential expansion to other home therapies. Future work will focus on three areas: efficiency, which includes model size, inference speed, energy use and consistency, which involves validating content and performing regression checks on new assets and impact, which will measure adherence and patient-reported outcomes in prospective studies.

ACKNOWLEDGMENT

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3. Upcoming Schedule (June 2026)

This month, the focus will shift to integrating the live cloud API and deploying the machine learning model onto the physical edge hardware.

- **Week 9:** Receive the trained Keras anomaly-detection model, convert it to TensorFlow Lite (TFLite) for reduced memory footprint, and deploy it for local inference on the Raspberry Pi.
- **Week 10:** Integrate the Vertex AI API (Google Workspace Pro subscription arriving this week). Connect the offline Python chart/subtitle pipeline to the Google Veo endpoint to generate the first live "Image-to-Video" animations.
- **Week 11:** Implement a local mock CAMARA QoD API endpoint using Python to simulate and test our 5G/6G bandwidth reservation logic prior to physical network testing.
- **Week 12:** Finalize end-to-end testing of the pipeline (anomaly detection → QoD reservation → video delivery) and prepare for Berlin Demo ,scheduled for end of June 2026.

4. Difficulties Encountered & Resolutions

- **Delay in Machine Learning Model Delivery:**
 - *Resolution:* The trained Keras model was delayed, preventing end-to-end edge deployment in May. To maintain project momentum, I utilized mock anomaly event triggers to test the orchestration flow and shifted my focus to finalizing the IEEE EMBS BSN conference paper. I am fully prepared to convert and deploy the model the moment it is provided.
- **Cloud API Access Restrictions:**
 - *Resolution:* We lacked the necessary Google Workspace Pro subscription to access the Vertex AI developer endpoints. To bypass this temporary blocker, I built all data decoupling modules locally (Matplotlib charts and .vtt subtitle generators). With the subscription confirmed to arrive this week, the pipeline is entirely ready to connect to the Veo API without further architectural changes.